RAG and Vector Databases - Introduction and Overview

Vector databases form the backbone of modern Retrieval-Augmented Generation systems, enabling efficient semantic search across vast document collections. A vector database is a specialized data store optimized for storing, indexing, and retrieving high-dimensional vector embeddings. Unlike traditional databases that store structured data in rows and columns, vector databases manage embeddings—numerical representations of text, images, or other data converted to high-dimensional vectors. The integration of RAG with vector databases creates a powerful synergy: embeddings capture semantic meaning, enabling searches based on conceptual similarity rather than keyword matching. Vector databases excel at similarity search, finding vectors closest to a query vector in the embedding space. This capability revolutionized information retrieval by enabling semantic understanding of content. Popular vector databases include Pinecone, Weaviate, Qdrant, Milvus, FAISS, Chroma, and Vespa. Each offers distinct features regarding scalability, performance characteristics, and operational requirements. The combination of embedding models, vector databases, and language models creates systems that retrieve semantically relevant information and generate contextually informed responses.